Cloud Computing Interview Questions and Answers (1–27)

These answers use simple English and are designed for B.Tech students preparing for fresher-level technical interviews. Each answer contains at least six points wherever appropriate, with examples to make the concepts easier to understand.

1. What is cloud computing? Explain the advantages of cloud computing.

Definition: Cloud computing is the delivery of computing services, such as servers, storage, databases, networking, and software, over the internet.

Instead of buying and maintaining physical computers and servers, we can use resources provided by cloud companies whenever we need them.

Advantages of cloud computing:

1. Cost saving: We do not need to purchase expensive physical servers.
2. Scalability: We can increase or decrease computing resources based on our needs.
3. Accessibility: We can access cloud services from anywhere using the internet.
4. High availability: Cloud providers offer services designed to remain available, depending on the configuration.
5. Backup and recovery: We can store backups and restore data after failures.
6. Fast deployment: We can create servers and databases within minutes.
7. Maintenance: Cloud providers manage much of the physical infrastructure.
8. Pay-as-you-go: We usually pay according to the resources we use.

Example: Suppose we develop an AI application. Instead of buying a powerful computer, we can use a cloud server to train and deploy the model.

2. Who is a cloud engineer?

Definition: A cloud engineer is an IT professional who designs, builds, deploys, manages, and maintains cloud-based systems.

Responsibilities of a cloud engineer:

1. Cloud infrastructure: Creates and manages virtual machines, storage, and networks.
2. Deployment: Deploys applications on cloud platforms.

3. Security: Configures access permissions, firewalls, and security policies.
4. Monitoring: Checks server performance, availability, and errors.
5. Backup and recovery: Creates backup systems to protect important data.
6. Cost management: Monitors cloud usage and reduces unnecessary expenses.
7. Automation: Uses scripts and tools to automate repetitive tasks.
8. Troubleshooting: Identifies and fixes cloud infrastructure problems.

Example: A cloud engineer can deploy a Python application on AWS, configure its network, set up monitoring, and ensure that only authorized users can access it.

Important skills: Linux, networking, AWS/Azure/GCP, scripting, security, Git, and infrastructure automation.

3. What is a cloud? List the different types of cloud.

Definition: A cloud is a collection of computing resources available through a network, usually the internet, that can be accessed when required.

In cloud computing, the word cloud refers to computing infrastructure and services rather than a physical cloud.

There are four commonly discussed cloud deployment models:

1. Public cloud: Infrastructure is provided by a third-party company and shared among multiple customers. Example: AWS.
2. Private cloud: Infrastructure is dedicated to one organization. Example: A company operating its own private cloud.
3. Hybrid cloud: Combines private infrastructure with public cloud services. Example: A bank keeps sensitive records in a private environment and uses public cloud resources for other applications.
4. Community cloud: Infrastructure is shared by organizations with similar requirements, such as government departments or healthcare institutions.

Example: A company may use a private cloud for confidential information and a public cloud for its website.

Interview point: Public, private, hybrid, and community are deployment models. IaaS, PaaS, and SaaS are service models.

4. Explain the different types of cloud service models.

Cloud service models describe the level of computing service provided by a cloud company and the responsibilities handled by the customer.

The three main service models are:

1. IaaS (Infrastructure as a Service): Provides virtual machines, networking, and storage. Example: AWS EC2.
2. PaaS (Platform as a Service): Provides a platform for developing and deploying applications without managing most underlying infrastructure. Example: Google App Engine.
3. SaaS (Software as a Service): Provides ready-to-use software over the internet. Example: Microsoft 365.
4. FaaS (Function as a Service): Runs individual functions when triggered. Example: AWS Lambda.
5. Customer responsibility: In IaaS, customers manage more of the software environment; in SaaS, the provider manages most of it.
6. Management level: IaaS gives greater control, PaaS reduces infrastructure management, and SaaS requires the least technical administration.
7. Common purpose: IaaS is useful for infrastructure, PaaS for application development, and SaaS for using complete software applications.

Example: If you rent a virtual server, you use IaaS. If you deploy code on a managed application platform, you use PaaS. If you use Gmail through a browser, you use SaaS.

5. Explain briefly about IaaS, PaaS, and SaaS.

[The Ultimate Guide to Cloud Infrastructure and Cloud Networking - UptimeRobot Knowledge Hub]

1. IaaS — Infrastructure as a Service

- Provides virtual machines, storage, and networking.
- The customer manages the operating system, applications, and data.
- The provider manages the physical infrastructure.
- Offers high flexibility and control.

Example: AWS EC2, Azure Virtual Machines.

[PPT - Cloud Computing Benefits and Challenges PowerPoint Presentation, free download - ID:4835914]

2. PaaS — Platform as a Service

- Provides a managed environment for application development and deployment.
- Developers focus mainly on code and application logic.
- The provider manages the operating system and much of the runtime infrastructure.
- Reduces infrastructure maintenance.

Example: Google App Engine, Azure App Service.

Software as a service cloud system on the laptop screen with floating shapes around. Networking, graph, security, data servers, management system

3. SaaS — Software as a Service

- Provides complete software through the internet.
- Users generally access it through a browser or application.
- The provider manages the application and its infrastructure.
- Users do not need to install or maintain the underlying servers.

Example: Gmail, Microsoft 365, Salesforce.

6. Explain the benefits of IaaS, PaaS, and SaaS.

Benefits of IaaS

1. Reduces the need to purchase physical servers.
2. Allows users to select computing resources based on their needs.
3. Supports quick creation and removal of virtual machines.
4. Provides control over operating systems and installed software.
5. Makes it easier to scale infrastructure.
6. Supports different operating systems and workloads.

Example: A company creates Linux virtual machines on AWS instead of buying physical servers.

Benefits of PaaS

1. Developers can focus on writing code.
2. Reduces the need to manage operating systems.
3. Speeds up application development.
4. Simplifies application deployment.
5. Offers managed runtime environments.
6. Reduces routine infrastructure maintenance.

Example: A developer deploys a Python web application using Azure App Service.

Benefits of SaaS

1. Users can access software through the internet.
2. Usually requires little local installation.
3. The provider handles software updates.
4. Reduces the need to maintain application servers.
5. Supports access from different devices.

6. Often provides subscription-based pricing.

Example: Students use Google Docs to create documents without managing the software's servers.

7. What is Function as a Service (FaaS)?

Definition: Function as a Service (FaaS) is a cloud computing model that allows developers to run individual functions in response to events without managing the underlying servers.

It is commonly associated with serverless computing.

1. Function-based execution: Developers write small units of code called functions.
2. Event-driven: Functions run when an event occurs, such as an HTTP request or file upload.
3. Server management: The cloud provider manages the underlying servers.
4. Automatic scaling: The platform can scale function execution according to demand.
5. Usage-based billing: Charges are commonly based on execution time and other resource usage, depending on the service.
6. Short-lived tasks: Functions are often used for small, event-driven operations rather than continuously running applications.
7. Examples: AWS Lambda, Azure Functions, and Google Cloud Run functions.

Example: When a user uploads an image, a cloud function automatically resizes the image and stores the processed version.

Advantages: Less server management, automatic scaling, and potentially lower costs for intermittent workloads.

Limitations: Execution limits, cold-start delays in some configurations, and challenges with long-running tasks.

8. Define the various cloud storage levels in cloud computing.

Cloud storage means storing data on cloud infrastructure instead of relying only on local storage devices. The main storage types are:

Cloud storage: file storage, block storage, or object storage?

1. Object storage

- Stores data as objects with data, metadata, and a unique identifier.
- Suitable for large amounts of unstructured data.

- Common uses include images, videos, backups, and datasets.
- Example: Amazon S3, Azure Blob Storage, Google Cloud Storage.

Block Storage | Cloud Product | Samsung SDS Vietnam

2. Block storage

- Divides data into fixed-size blocks.
- Commonly used as disks for virtual machines and databases.
- Provides low-latency access for many workloads.
- Example: Amazon EBS, Azure Managed Disks, Google Persistent Disk.

NFSv4.x pseudo-file system | Dell PowerScale: OneFS NFS Design Considerations and Best Practices | Dell Technologies Info Hub

3. File storage

- Organizes data into files and folders.
- Uses a hierarchical directory structure.
- Useful for shared documents and applications that require a file system.
- Example: Amazon EFS, Azure Files, Google Cloud Filestore.

4. Archive storage: Designed for data that is rarely accessed but must be retained for a long time. Example: Amazon S3 Glacier storage classes.

5. Backup storage: Stores copies of important data to help recover it after accidental deletion or system failure.

6. Storage tiers: Providers may offer different cost and access tiers, such as frequent-access, infrequent-access, and archive tiers.

Interview point: Object, block, and file storage are storage types. Hot, cool, and archive are access or cost tiers offered by some cloud storage services. Do not confuse the two classifications.

9. What are the key features of cloud computing?

1. On-demand self-service: Users can create resources whenever needed without waiting for manual provider intervention.
2. Broad network access: Services can be accessed through standard networks using computers, phones, and other devices.
3. Resource pooling: Providers share physical resources among multiple customers while logically separating their workloads.
4. Rapid elasticity: Resources can increase or decrease quickly as demand changes.
5. Measured service: Resource usage is monitored for billing and management.

6. Scalability: Systems can support growing workloads by adding resources.
7. High availability options: Providers offer features that help applications remain available during certain failures.
8. Automation: Cloud resources can be created and managed using scripts, APIs, and templates.

Example: An online shopping website can increase server capacity during a festival sale and reduce it when traffic decreases.

10. Explain public, private, and hybrid cloud.

1. Public cloud

- Services are provided by a third-party cloud provider.
- Infrastructure is shared among multiple customers.
- Each customer has logically separated resources.
- Usually offers flexible pricing and quick setup.
- Examples: AWS, Microsoft Azure, Google Cloud.

Apa Itu Private Cloud: Pengertian, Cara Kerja, dan Manfaatnya

2. Private cloud

- Cloud infrastructure is dedicated to one organization.
- It can run in the organization's own data centre or a hosted environment.
- Provides greater control over configuration and access.
- Requires more management and planning.
- Example: A bank operating a private cloud using OpenStack.

Hybrid Cloud Architecture This diagram will illustrate the core components of a hybrid cloud: your on-premises data center, a public cloud provider, and the secure connection between them. Key… | Sukhen Tiwari

3. Hybrid cloud

- Combines private cloud infrastructure with public cloud services.
- Allows data and applications to work across both environments when properly integrated.
- Can keep sensitive workloads in a controlled environment.
- Can use public cloud resources for changing demand.
- Example: A company keeps confidential records privately and hosts its public website on AWS.

Important distinction: Using a private network connection to a public cloud does not, by itself, make the setup a hybrid cloud. Hybrid cloud involves integration between distinct environments.

11. What are the benefits and limitations of private and public clouds?

Public cloud

Benefits

1. Lower initial cost: No need to purchase a complete physical data centre.
2. Easy scalability: Resources can be added when demand increases.
3. Fast deployment: Services can be created within minutes.
4. Global reach: Providers offer infrastructure across multiple regions.
5. Managed services: Providers offer databases, AI tools, monitoring, and other services.
6. Flexible pricing: Many services charge according to consumption.

Limitations

1. Costs can become high if resources are poorly managed.
2. Customers have less control over the underlying physical infrastructure.
3. Internet or network connectivity can affect access.
4. Configuration mistakes can create security risks.
5. Some workloads may face regulatory or data-residency restrictions.
6. Moving applications between providers can require significant effort.

Example: A startup uses AWS to launch its application without buying servers.

Private cloud

Benefits

1. Greater control over infrastructure.
2. Dedicated environment for one organization.
3. Custom security and network configurations.
4. Can support strict compliance requirements.
5. Resources can be customized for specific workloads.
6. Offers more control over where infrastructure is located.

Limitations

1. High initial setup cost in many deployments.
2. Requires skilled staff to manage infrastructure.
3. The organization is responsible for more maintenance.
4. Scaling may require purchasing additional hardware.
5. Hardware failures require suitable recovery planning.
6. Capacity can be wasted when resources are underused.

Example: A financial institution operates its own private cloud to meet internal security and regulatory requirements.

12. What are the benefits of a hybrid cloud?

1. Flexibility: Organizations can choose where to run different applications.
2. Sensitive data control: Confidential workloads can remain in a private environment.
3. Scalability: Public cloud resources can support sudden increases in demand.
4. Cost optimization: Organizations can choose suitable environments based on workload and total cost.
5. Disaster recovery: Public cloud services can be used as part of a recovery strategy.
6. Gradual migration: Companies can move applications to the public cloud in stages.
7. Business continuity: Properly designed systems can continue operating during certain infrastructure failures.
8. Workload separation: Applications with different security and performance requirements can run in suitable environments.

Example: A college stores sensitive student records in a controlled private environment but uses public cloud services to host its website and learning portal.

Limitation to remember: Hybrid cloud can be complex because networking, identity management, security, monitoring, and data synchronization must work across environments.

13. Who are the cloud service providers in a cloud ecosystem?

Definition: Cloud service providers (CSPs) are companies that offer computing infrastructure, platforms, storage, databases, networking, and software services over a network.

Important providers include:

1. Amazon Web Services (AWS): Offers EC2, S3, Lambda, RDS, and many other services.
2. Microsoft Azure: Offers Azure Virtual Machines, Azure App Service, Azure SQL Database, and Azure Functions.
3. Google Cloud Platform (GCP): Offers Compute Engine, Cloud Storage, BigQuery, and Vertex AI.
4. Oracle Cloud Infrastructure (OCI): Offers cloud computing, databases, and enterprise infrastructure.
5. IBM Cloud: Provides enterprise cloud, AI, and hybrid-cloud solutions.
6. Alibaba Cloud: Provides cloud infrastructure and services, particularly across Asian markets.
7. Salesforce: Known for cloud-based CRM software and business applications.

Example: If a company wants to host its website on virtual servers, it can choose AWS EC2, Azure Virtual Machines, or Google Compute Engine.

Interview point: AWS, Azure, and GCP are commonly compared because they provide broad infrastructure, platform, database, networking, and AI services.

14. What is AWS, and why is it so popular?

Definition: Amazon Web Services (AWS) is a cloud computing platform provided by Amazon. It offers services for computing, storage, databases, networking, security, analytics, and artificial intelligence.

1. AWS allows businesses to run applications without purchasing all the underlying physical infrastructure.
2. It provides computing services such as Amazon EC2.
3. It provides storage services such as Amazon S3.
4. It offers managed database services such as Amazon RDS.
5. It provides serverless computing through AWS Lambda.
6. It offers infrastructure in multiple geographic regions.
7. It supports many programming languages, operating systems, and development tools.
8. It offers usage-based pricing for many services.

Example: A developer can deploy a Python application on an EC2 instance, store uploaded images in S3, and use RDS to store application data.

15. Explain the reasons why AWS is so popular.

1. Wide range of services: AWS offers services for computing, storage, databases, security, AI, and analytics.
2. Scalability: Resources can be adjusted as application demand changes.
3. Global infrastructure: AWS operates regions and availability zones in different parts of the world.
4. Flexible pricing: Many services support pay-as-you-go usage.
5. Reliability options: Multiple availability zones and backup services can help build resilient systems.
6. Security tools: AWS provides identity, access, encryption, logging, and security monitoring features.
7. Developer support: It offers APIs, SDKs, command-line tools, and automation capabilities.
8. Ecosystem: There are many learning resources, third-party integrations, and job opportunities involving AWS.

Example: An e-commerce company can use EC2 for computing, S3 for images, CloudFront for content delivery, and RDS for its database.

Interview caution: AWS does not automatically make an application secure, highly available, or inexpensive. Those outcomes depend on architecture and configuration.

16. What are AWS services?

AWS services are cloud-based tools that help users build, deploy, store, secure, and manage applications.

The main AWS services are:

Service	Purpose	Example use
Amazon EC2	Virtual servers	Host a web application
Amazon S3	Object storage	Store images, files, and backups
Amazon RDS	Managed relational databases	Store customer records
AWS Lambda	Serverless functions	Process uploaded files
Amazon VPC	Virtual networking	Create an isolated cloud network
AWS IAM	Identity and access management	Control user permissions
Amazon CloudWatch	Monitoring and logs	Monitor CPU usage and errors
Amazon CloudFront	Content delivery network	Deliver website content faster

Amazon Route 53	DNS and domain routing	Route users to applications
Amazon EBS	Block storage	Attach storage volumes to EC2
Amazon EKS	Managed Kubernetes	Run containerized applications
Amazon SageMaker	AI/ML development	Train and deploy machine learning models

Six important points to remember:

1. AWS provides computing services to run applications.
2. It provides storage services to save data.
3. It provides database services to organize and retrieve data.
4. It provides networking services to connect resources.
5. It provides security services to manage permissions and protect workloads.
6. It provides monitoring, analytics, and AI/ML services.

Example: For an AI project, you might use EC2 to run the application, S3 to store datasets, IAM to control access, and CloudWatch to monitor performance.

17. What is Microsoft Azure Cloud? What is Azure famous for?

Definition: Microsoft Azure is a cloud computing platform developed by Microsoft. It provides infrastructure, application hosting, databases, networking, security, analytics, and AI services.

1. Virtual machines: Azure provides virtual servers through Azure Virtual Machines.
2. Application hosting: Azure App Service supports hosting web applications and APIs.
3. Storage: Azure Blob Storage stores files, images, videos, and other objects.
4. Databases: Azure SQL Database provides a managed relational database service.
5. AI services: Azure provides AI and machine learning capabilities, including Azure AI Foundry and Azure Machine Learning.
6. Identity management: Microsoft Entra ID supports identity and access management.
7. Hybrid cloud: Azure offers tools for connecting on-premises infrastructure with cloud services.

8. Microsoft integration: Azure works closely with Microsoft technologies, including Windows Server, Microsoft 365, and other enterprise systems.

What is Azure famous for?

Azure is particularly known for its integration with Microsoft's enterprise ecosystem, hybrid-cloud capabilities, and enterprise identity and management tools. It also offers a broad range of general-purpose cloud services.

Example: A company using Windows Server and Microsoft Entra ID can use Azure to host its applications and manage user access.

18. What are the main services offered by Azure?

Azure service	Purpose	Example
Azure Virtual Machines	Virtual computing	Run Windows or Linux servers
Azure Blob Storage	Object storage	Store images and backups
Azure SQL Database	Managed relational database	Store business records
Azure App Service	Web application hosting	Deploy a Python web API
Azure Functions	Serverless computing	Process events automatically
Azure Virtual Network	Private cloud networking	Connect cloud resources
Microsoft Entra ID	Identity and access	Manage user authentication

Azure Monitor	Monitoring	Track application performance
Azure Kubernetes Service (AKS)	Managed Kubernetes	Deploy containerized applications
Azure Machine Learning	ML development and deployment	Train and manage ML models

Key points:

1. Azure provides infrastructure for running applications.
2. It offers storage for files and application data.
3. It provides managed database services.
4. It supports serverless computing and application hosting.
5. It provides security, identity, and networking tools.
6. It offers analytics, AI, and machine learning services.

Example: A developer can host a Python API on Azure App Service, store images in Blob Storage, and save application data in Azure SQL Database.

19. What is a data centre?

Definition: A data centre is a physical facility that contains servers, storage systems, networking equipment, power systems, and cooling systems used to process and store data.

Important points:

1. Servers: Run applications, websites, databases, and other software.
2. Storage systems: Store files, databases, and backups.
3. Networking equipment: Connects servers and links the facility to external networks.
4. Power supply: Uses electrical systems and often backup power to support operations.
5. Cooling systems: Remove heat generated by servers and networking equipment.
6. Physical security: Uses measures such as access controls and surveillance to protect equipment.
7. Reliability: May include redundant power, network connections, and hardware to reduce downtime.
8. Cloud foundation: Cloud providers operate data centres to deliver cloud services.

Example: When you create a virtual machine on AWS, the underlying physical server runs in an AWS data centre. You manage the virtual machine, while AWS manages the physical facility and hardware.

Difference between a data centre and cloud computing:

- A data centre is a physical facility.
- Cloud computing is a way of providing and accessing computing resources as services.
- A cloud provider may operate many data centres across different regions.

20. What is Google Cloud Platform (GCP)?

Definition: Google Cloud Platform, commonly called Google Cloud, is a cloud computing platform provided by Google. It offers computing, storage, databases, networking, analytics, and AI services.

1. Computing: Google Compute Engine provides virtual machines.
2. Storage: Google Cloud Storage stores objects such as images, videos, and datasets.
3. Databases: Google Cloud offers managed database services such as Cloud SQL and Firestore.
4. Data analytics: BigQuery supports large-scale data analysis using SQL.
5. AI and machine learning: Vertex AI supports developing, training, and deploying ML models.
6. Serverless computing: Cloud Run runs containerized applications without requiring users to manage servers.
7. Networking: Google Cloud provides virtual networks, load balancing, and connectivity services.
8. Container management: Google Kubernetes Engine (GKE) provides managed Kubernetes.

Example: A data scientist can store a dataset in Cloud Storage, analyze it using BigQuery, train a model using Vertex AI, and deploy an application using Cloud Run.

21. What are the advantages of GCP?

1. Data analytics: BigQuery is useful for analyzing large datasets.
2. AI and ML: Vertex AI supports machine learning development and deployment.
3. Scalability: Resources can be adjusted according to application requirements.
4. Global infrastructure: Google Cloud offers regions and zones across multiple geographic locations.
5. Container support: GKE provides managed Kubernetes capabilities.
6. Serverless deployment: Cloud Run allows developers to deploy containerized applications without managing servers.

7. Managed services: Many infrastructure and database maintenance tasks are handled by Google.
8. Integration with Google tools: Google Cloud integrates with Google's data, analytics, and AI ecosystem.

Example: An AI company can use BigQuery to analyze millions of records and Vertex AI to train and deploy a prediction model.

Limitation: GCP still requires knowledge of cloud architecture, security, pricing, and service configuration. Its benefits do not eliminate the need for technical skills.

22. Explain the main differences between AWS, Azure, and GCP.

AWS, Microsoft Azure, and Google Cloud are three major cloud platforms. They provide similar core services, but differ in their ecosystems, strengths, tools, and pricing options.

Feature	AWS	Microsoft Azure	Google Cloud
Provider	Amazon	Microsoft	Google
Virtual machines	EC2	Azure Virtual Machines	Compute Engine
Object storage	S3	Blob Storage	Cloud Storage
Managed SQL database	Amazon RDS	Azure SQL Database	Cloud SQL
Serverless computing	Lambda	Azure Functions	Cloud Run functions
Kubernetes	EKS	AKS	GKE

AI/ML platform	SageMaker	Azure Machine Learning	Vertex AI
Data analytics	Athena, Redshift	Microsoft Fabric, Synapse Analytics	BigQuery
Common strength	Broad cloud service ecosystem	Microsoft enterprise integration and hybrid cloud	Data analytics, AI/ML, and Kubernetes

Important differences:

1. AWS: Offers a broad range of cloud services and infrastructure options.
2. Azure: Has strong integration with Microsoft's enterprise products and identity systems.
3. GCP: Is well known for BigQuery, data analytics, and its Kubernetes ecosystem.
4. Pricing: All three offer multiple pricing models. Actual costs depend on resource usage, region, and configuration.
5. Learning: All three require knowledge of networking, security, computing, and cloud architecture.
6. Career opportunities: All three are relevant to cloud engineering, DevOps, and AI/ML roles.

Example: If you want to host a Python application, you could use EC2 on AWS, Azure App Service on Azure, or Cloud Run on Google Cloud.

Interview conclusion: There is no universally best cloud provider. The right choice depends on the application, budget, existing technology, security requirements, and team expertise.

23. Compare AWS EC2, Azure VM, and Google Compute Engine.

These three services provide virtual machines (VMs). A virtual machine behaves like a computer with its own operating system, CPU, memory, storage, and network configuration.

AWS EC2 (Elastic Compute Cloud) Logo PNG, AI, EPS, CDR, PDF, SVG - iconLogoVector

1. AWS EC2

- Full name: Amazon Elastic Compute Cloud.
- Provides resizable virtual computing capacity.

- Supports Linux and Windows instances.
- Allows users to choose instance types based on CPU, memory, and workload.
- Commonly integrates with EBS, VPC, IAM, and CloudWatch.
- Example: Hosting a Python application on an EC2 instance.

[Azure Virtual Machines | Learn the Basic Advantages and Architecture]

2. Azure Virtual Machines

- Provides on-demand virtual machines on Azure.
- Supports Windows and Linux.
- Offers different VM sizes for different workloads.
- Integrates with Azure Managed Disks and Virtual Network.
- Supports Azure Monitor and Microsoft Entra ID integration.
- Example: Hosting a Windows-based business application on Azure.

[Dịch vụ Google Compute Engine - Gimasys]

3. Google Compute Engine

- Provides virtual machines on Google Cloud.
- Supports Linux and Windows workloads.
- Offers predefined and customizable machine configurations.
- Integrates with Google Cloud Storage and Virtual Private Cloud.
- Supports monitoring through Google Cloud's operations tools.
- Example: Running a machine learning API on a Compute Engine VM.

Main comparison:

Feature	EC2	Azure VM	Compute Engine
Provider	AWS	Microsoft	Google
Main purpose	Virtual computing	Virtual computing	Virtual computing
VM configuration	Instance types	VM sizes	Machine types

Networking	Amazon VPC	Azure VNet	Google Cloud VPC
Attached disk service	EBS	Azure Managed Disks	Persistent Disk / Hyperdisk, depending on use
Monitoring	CloudWatch	Azure Monitor	Google Cloud Observability tools

Interview point: These services perform the same basic function. Their interfaces, configuration options, integrations, and pricing differ. None is automatically the fastest or cheapest for every workload.

24. What is virtualization?

Definition: Virtualization is a technology that creates virtual versions of computing resources, such as servers, operating systems, storage, and networks, using software.

It allows one physical computer to run multiple isolated virtual machines.

Important points:

1. Resource sharing: Multiple virtual machines can use the resources of one physical server.
2. Virtual machines: Each VM can run its own operating system and applications.
3. Isolation: Problems inside one VM are generally separated from other VMs, subject to the security of the virtualization platform.
4. Resource allocation: CPU, memory, storage, and networking resources can be allocated to VMs.
5. Cost reduction: Organizations can reduce the number of physical servers they need.
6. Easy testing: Developers can create separate environments for testing different operating systems.
7. Flexibility: VMs can often be created, copied, moved, or deleted more easily than physical machines.
8. Cloud foundation: Virtualization is an important technology used in many cloud environments, although modern cloud platforms also use containers and other technologies.

Example: Suppose a physical server has 32 GB RAM. Using a hypervisor, an administrator can create several VMs and allocate memory and CPU resources to each one according to their requirements.

Physical server

CPU · RAM · Storage

Hypervisor

VM 1

Linux

VM 2

Windows

VM 3

Linux

Advantages: Better hardware utilization, isolation, flexible testing, and simpler infrastructure management.

Limitations: Virtualization introduces some overhead, requires resources for every VM, and needs proper configuration to maintain security and performance.

Real-world usage: Data centres, cloud computing, software testing, virtual desktops, and server consolidation.

25. What is a hypervisor?

Definition: A hypervisor is software, firmware, or a hardware-assisted virtualization layer that creates and manages virtual machines by allocating physical resources to them.

It is also called a virtual machine monitor (VMM).

Important points:

1. Creates VMs: It enables virtual machines to be created on a physical system.
2. Allocates resources: It assigns CPU time, memory, and virtual devices to VMs.
3. Manages execution: It coordinates the execution of virtual machines on the underlying hardware.
4. Provides isolation: It helps keep VM environments separate.
5. Manages virtual hardware: It presents virtual processors, disks, network adapters, and other devices.
6. Supports multiple operating systems: Different VMs can run different supported operating systems on one physical server.
7. Improves utilization: It allows physical resources to be shared among workloads.

8. Supports cloud infrastructure: Hypervisors are used in many virtual-machine-based cloud environments.

Examples of hypervisors:

- VMware ESXi
- Microsoft Hyper-V
- KVM
- Oracle VirtualBox
- VMware Workstation

Example: If you run Kali Linux in VirtualBox on your Windows laptop, VirtualBox manages the virtual machine and its allocated resources.

Interview distinction: Virtualization is the technology; a hypervisor is one of the components used to implement it.

26. What are the two types of hypervisors?

There are two main types of hypervisors: Type I and Type II.

1. Type I is also called a bare-metal hypervisor. It runs directly on the physical hardware.
2. Type II is also called a hosted hypervisor. It runs as software on top of a host operating system.
3. Type I manages VMs without requiring a general-purpose host operating system beneath the hypervisor.
4. Type II depends on the host OS for many hardware and device interactions.
5. Type I is common in data centres and enterprise virtualization environments.
6. Type II is common for learning and development on personal computers.
7. Performance and overhead differ: Type I often has less host-OS overhead, but actual performance depends on the implementation and workload.
8. Examples differ: VMware ESXi and Hyper-V's hypervisor architecture are Type I examples; VirtualBox and VMware Workstation are Type II examples in their usual hosted configurations.

Example: A company can use VMware ESXi on a physical server to host many production VMs. A student can use VirtualBox on a Windows laptop to practice Kali Linux.

27. Explain the difference between Type I and Type II hypervisors.

Feature	Type I hypervisor	Type II hypervisor

Other name	Bare-metal	Hosted
Runs on	Physical hardware	Host operating system
Host OS beneath hypervisor	No general-purpose host OS is required	Yes
Common use	Data centres and production servers	Learning, testing, and desktop development
Performance	Often has lower host-OS overhead	Adds host-OS overhead
Resource management	Manages hardware resources for VMs	Relies on the host OS for many operations
Setup	Often installed as a virtualization platform	Usually installed like a normal application
Examples	VMware ESXi, Xen, Hyper-V	Oracle VirtualBox, VMware Workstation

Type I hypervisor — six points

1. Runs directly on physical hardware.
2. Manages virtual machines and their hardware resources.
3. Is commonly used in enterprise and cloud infrastructure.
4. Usually has less overhead from a general-purpose host OS.
5. Can support many VMs, depending on available hardware and configuration.
6. Requires appropriate setup and administration.

Type II hypervisor — six points

1. Runs on an existing operating system such as Windows or Linux.
2. Is installed and used like an application.
3. Is useful for learning and testing different operating systems.
4. Shares physical resources with the host OS and other applications.
5. Can have additional performance overhead.
6. Is generally convenient for desktop development and lab environments.

Simple example:

- Type I: A company installs ESXi on a physical server and runs multiple production VMs.
- Type II: A student installs VirtualBox on a Windows laptop and runs Kali Linux inside a VM.

Interview-ready conclusion: The primary difference is where the hypervisor runs. Type I runs directly on the hardware, while Type II runs on top of a host operating system.